University of Puerto Rico Mayagüez Campus

Electrical and Computer Engineering Department

AI Agent Architecture of Artemis IIs Mission

Prepared by:
Jael A. Gonzalez De Arce
Néstor O. Santos Torres
Jayson D. Perez Ramirez

Date:
April 29, 2026

Table of Contents

I. Introduction

The Artemis II mission represents a major milestone in human space exploration as the first crewed mission of NASA's Artemis program, designed to execute a lunar flyby using the Orion spacecraft and Space Launch System (SLS) [1]. Although not explicitly categorized as an artificial intelligence system, Artemis II can be interpreted as a distributed intelligent architecture composed of interacting agents, each responsible for perception, reasoning, and action. This perspective aligns with established AI frameworks, where intelligence emerges from coordinated agent behavior across multiple levels of abstraction [2].

This report models Artemis II as a hierarchical multi-agent system, identifying agent types, their functional roles, system architecture, and human involvement, while also evaluating system robustness through a failure scenario.

II. Agent Identification and Classification

The Artemis II system can be decomposed into several interacting agents, each fulfilling distinct roles within the mission architecture.

A. Guidance, Navigation, and Control (GNC) System: Model-Based Agent

The Guidance, Navigation, and Control system operates as a model-based agent, maintaining an internal representation of the spacecraft's state, including position, velocity, and orientation. It continuously updates this model using sensor data such as inertial measurement units and star trackers, allowing it to make informed control decisions in dynamic conditions [3].

B. Flight Control System (FCS): Simple Reflex Agent

The Flight Control System functions as a simple reflex agent, executing immediate control actions based on real-time sensor inputs. Its primary responsibility is maintaining spacecraft stability through rapid adjustments, such as thruster firings, without relying on long-term planning or internal modeling [2].

C. Mission Planning System: Goal-Based Agent

The Mission Planning System acts as a goal-based agent, responsible for determining trajectories, mission phases, and maneuver sequences required to achieve mission objectives such as lunar flyby and safe return to Earth. It evaluates future states and selects actions aligned with mission goals [1].

D. Fault Detection, Isolation, and Recovery (FDIR) System: Utiliy-Based Agent

The Fault Detection, Isolation, and Recovery system can be classified as a utility-based agent, as it evaluates multiple possible recovery strategies during anomalies and selects the one that maximizes mission safety and performance under uncertainty [4].

E. Human Agents (Astronauts and Ground Control)

Astronauts and mission control personnel function as human agents, capable of adaptive reasoning, contextual understanding, and decision-making in unforeseen scenarios. They can supervise automated systems and intervene when necessary, providing an additional layer of intelligence and flexibility [5].

III. Functional Decomposition

The Artemis II architecture can be organized into hierarchical functional layers, each responsible for different aspects of system intelligence.

A. Low-Level Reactive Control Layer

The low-level layer consists of reactive control agents such as the Flight Control System, which handle real-time stabilization and actuation. These agents prioritize speed and reliability, responding instantly to environmental changes without complex reasoning [2].

B. Mid-Level State Estimation Layer

The mid-level layer includes model-based agents such as the GNC system. These agents integrate sensor data to maintain an accurate estimate of the spacecraft's state, enabling coordination between perception and control [3].

C. High-Level Planning Layer

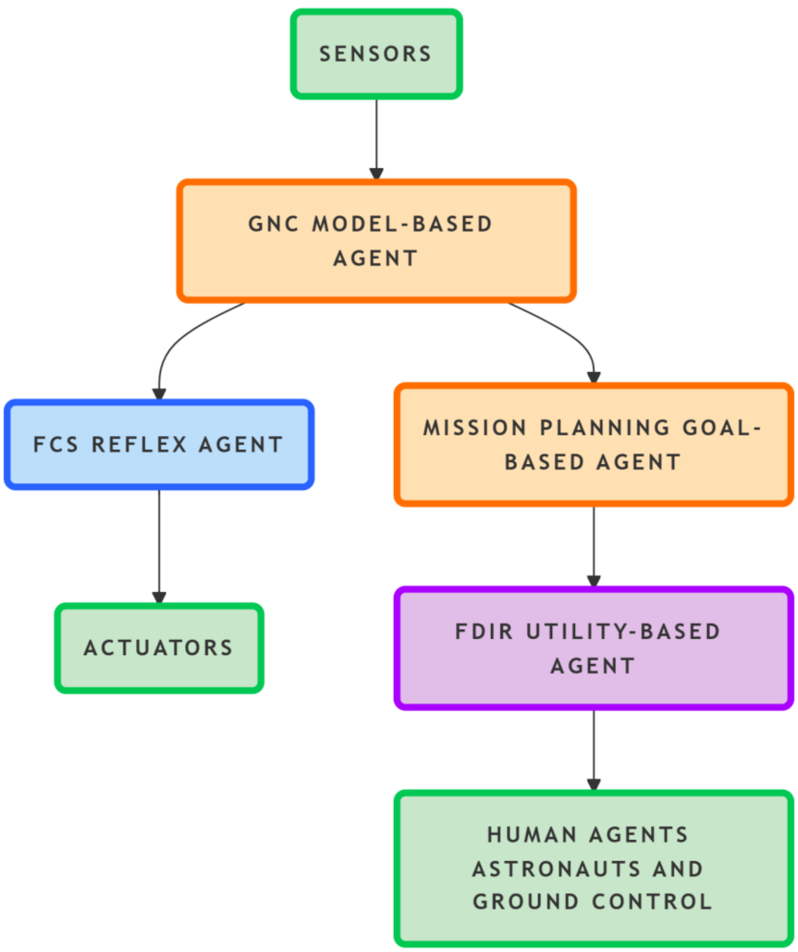
The high-level layer consists of goal-based and utility-based agents, including mission planning and FDIR systems. These agents perform trajectory optimization, anomaly evaluation, and long-term decision-making based on predicted outcomes [1], [4].

D. Human-in-the-Loop Layer

The highest layer incorporates human agents, including astronauts and ground control. This layer provides supervision, validation, and intervention capabilities, ensuring that the system can adapt to unexpected situations beyond automated reasoning [5].

IV. Interconnections and Architecture

The Artemis II system is characterized by continuous information exchange between agents, forming a tightly integrated architecture.



A. Sensor-to-Model Interaction

Sensor systems provide telemetry data to the GNC system, which processes and integrates this information into a coherent state estimate. This interaction forms the foundation of system awareness and enables accurate control decisions [3].

B. Control and Actuation Flow

The GNC system transmits control signals to the Flight Control System, which executes these commands through actuators such as thrusters. Feedback loops ensure that system behavior is continuously monitored and corrected in real time [2].

C. Planning and Feedback Integration

High-level agents, including mission planning and FDIR systems, receive state information and provide strategic directives. These directives influence both mid-level modeling and low-level control, creating a hierarchical feedback structure [4].

D. Human-System Interaction

Human agents interact with the system through telemetry and command interfaces. Ground control provides strategic decisions, while astronauts can directly intervene in onboard systems. Communication delays necessitate partial onboard autonomy, reinforcing the importance of distributed intelligence [5].

V. Perception–Decision–Action Mapping

The Artemis II system follows a structured perception–decision–action cycle, fundamental to intelligent systems.

A. Perception Components

Perception is achieved through onboard sensors, including inertial measurement units, star trackers, and telemetry systems. These components collect data about the spacecraft’s state and environment, forming the basis for decision-making [3].

B. State Estimation and Modeling

State estimation occurs within the GNC system, which fuses sensor inputs to maintain an accurate internal representation of the spacecraft's dynamics. This modeling capability is essential for predicting future states and guiding control actions [3].

C. Decision-Making Components

Decision-making is distributed across multiple agents. Low-level agents handle immediate responses, while higher-level systems determine trajectories and mission strategies based on goals and utility optimization [2].

D. Action and Execution Systems

Actions are executed through actuators such as thrusters and control mechanisms. These systems implement decisions made by higher-level agents, ensuring that the spacecraft follows the intended trajectory [1].

VI. Human-in-the-Loop Analysis

Human agents play a critical role in ensuring mission success and system adaptability.

A. Role of Astronauts

Astronauts act as onboard decision-makers capable of responding to unexpected conditions. They can manually control spacecraft systems and override automation when necessary, providing flexibility in uncertain environments [5].

B. Role of Ground Control

Ground control serves as a strategic oversight entity, responsible for mission planning, monitoring, and high-level decision-making. It provides guidance and support to both automated systems and astronauts [1].

C. Advantages and Limitations

The inclusion of human agents enhances adaptability and problem-solving capabilities. However, it introduces limitations such as communication delays and slower response times compared to automated systems, highlighting the need for balanced human-machine collaboration [4].

VII. Failure Scenario Analysis

A realistic failure scenario involves a malfunction in the star tracker system, which affects spacecraft orientation estimation.

A. Detection and Diagnosis

The GNC system detects inconsistencies between sensor inputs and internal models, triggering the FDIR system to analyze the anomaly and identify potential causes [3], [4].

B. System Response

The FDIR system evaluates alternative strategies, such as switching to redundant sensors or relying on inertial measurements. The Flight Control System continues operating using available data, maintaining stability [2].

C. Human Intervention

If uncertainty exceeds acceptable thresholds, astronauts or ground control may intervene to adjust mission parameters or initiate contingency procedures, ensuring mission safety [5].

D. Robustness and Limitations

This layered response demonstrates system robustness through redundancy and distributed decision-making. However, simultaneous failures or communication delays may limit system effectiveness, emphasizing the importance of resilient architecture design.

VIII. Conclusion

Reinterpreting Artemis II as a multi-agent system reveals the distributed nature of intelligence in complex aerospace missions. The integration of reflex, model-based, goal-based, utility-based, and human agents enables efficient operation across multiple levels of abstraction. This layered architecture balances autonomy and adaptability, demonstrating that intelligence emerges from the interaction of diverse agents rather than a single centralized system.

References

[1] NASA, "Artemis II: NASA's First Crewed Lunar Flyby in 50 Years" 2025. [Online]. Available: Artemis II: NASA's First Crewed Lunar Flyby in 50 Years - NASA

[2] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2021.

[3] NASA, "Orion Guidance, Navigation, and Control System," 2024.

[4] Y. Bar-Shalom, X. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*. Wiley, 2001.

[5] NASA, "Human Integration in Spaceflight Systems," 2023.

[6] J. Wertz and W. Larson, *Space Mission Analysis and Design*, 3rd ed. Microcosm Press, 1999.

[7] European Space Agency, "Autonomous Systems in Space Missions," 2022.

AI Acknowledgment

AI tools (e.g., ChatGPT) were used to support brainstorming and clarification. All final analysis and conclusions were independently verified by the author.